Họ và tên: Hoàng Việt Quang

MSSV: 066206005044

Mã học phần: 012012500106

a) LRU replacement

Bảng mô phỏng (H = hit, F = fault):

i	R_i	F1	F2	F3	F4	H/F
1	7	7	-	-	-	F
2	0	7	0	-	-	F
3	1	7	0	1	-	F
4	2	7	0	1	2	F
5	0	7	0	1	2	H
6	3	3	0	1	2	F
7	0	3	0	1	2	H
8	4	3	0	4	2	F
9	2	3	0	4	2	H
10	3	3	0	4	2	H
11	0	3	0	4	2	H
12	3	3	0	4	2	H
13	2	3	0	4	2	H

14	1	3	0	1	2	F
15	2	3	0	1	2	H
16	0	3	0	1	2	H
17	1	3	0	1	2	H
18	7	7	0	1	2	F
19	0	7	0	1	2	H
20	1	7	0	1	2	H

$$PF_{LRU} = 8$$

b) FIFO replacement

i	R_i	F1	F2	F3	F4	H/F
1	7	7	-	-	-	F
2	0	7	0	-	-	F
3	1	7	0	1	-	F
4	2	7	0	1	2	F
5	0	7	0	1	2	H
6	3	3	0	1	2	F
7	0	3	0	1	2	H
8	4	3	4	1	2	F

9	2	3	4	1	2	H
10	3	3	4	1	2	H
11	0	3	4	0	2	F
12	3	3	4	0	2	H
13	2	3	4	0	2	H
14	1	3	4	0	1	F
15	2	2	4	0	1	F
16	0	2	4	0	1	H
17	1	2	4	0	1	H
18	7	2	7	0	1	F
19	0	2	7	0	1	H
20	1	2	7	0	1	H

$$PF_{FIFO} = 10$$

c) Optimal replacement

i	R_i	F1	F2	F3	F4	H/F
1	7	7	-	-	-	F
2	0	7	0	-	-	F
3	1	7	0	1	-	F

4	2	7	0	1	2	F
5	0	7	0	1	2	H
6	3	3	0	1	2	F
7	0	3	0	1	2	H
8	4	3	0	4	2	F
9	2	3	0	4	2	H
10	3	3	0	4	2	H
11	0	3	0	4	2	H
12	3	3	0	4	2	H
13	2	3	0	4	2	H
14	1	1	0	4	2	F
15	2	1	0	4	2	H
16	0	1	0	4	2	H
17	1	1	0	4	2	H
18	7	1	0	7	2	F
19	0	1	0	7	2	H
20	1	1	0	7	2	H

$$PF_{OPT} = 8$$

⇨ $PF_{LRU} = 8, PF_{FIFO} = 10, PF_{OPT} = 8$